

The sample output of `gauss2rv` shown here is produced with the commands

```
>> xy=gauss2rv(3,3,5,1,0.5,500);
>> plot(xy(1,:),xy(2,:),'.');

```

We observe that the center of the cloud is $(\mu_X, \mu_Y) = (3, 5)$. In addition, we note that the X and Y axes are scaled differently because $\sigma_X = 3$ and $\sigma_Y = 1$.

We observe that this example with $\rho_{X,Y} = 0.5$ shows random variables that are less correlated than the examples in Figure 5.5 with $|\rho| = 0.9$.

We note that bivariate Gaussian random variables are a special case of n -dimensional Gaussian random vectors, which are introduced in Chapter 8. Based on linear algebra techniques, Chapter 8 introduces the `gaussvector` function to generate samples of Gaussian random vectors that generalizes `gauss2rv` to n dimensions.

Beyond bivariate Gaussian pairs, there exist a variety of techniques for generating sample values of pairs of continuous random variables of specific types. A basic approach is to generate X based on the marginal PDF $f_X(x)$ and then generate Y using a conditional probability model that depends on the value of X . Conditional probability models and MATLAB techniques that employ these models are the subject of Chapter 7.

Problems

Difficulty: ● Easy ■ Moderate ◆ Difficult ◆◆ Experts Only

5.1.1 ● Random variables X and Y have the joint CDF

$$F_{X,Y}(x,y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}) & x \geq 0; \\ & y \geq 0, \\ 0 & \text{ow.} \end{cases}$$

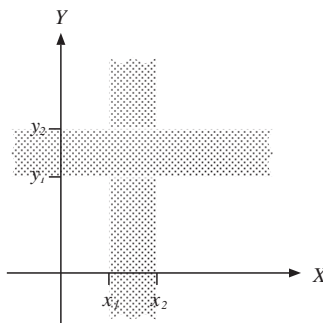
- (a) What is $P[X \leq 2, Y \leq 3]$?
- (b) What is the marginal CDF, $F_X(x)$?
- (c) What is the marginal CDF, $F_Y(y)$?

5.1.2 ● Express the following extreme values of $F_{X,Y}(x,y)$ in terms of the marginal cumulative distribution functions $F_X(x)$ and $F_Y(y)$.

- (a) $F_{X,Y}(x, -\infty)$
- (b) $F_{X,Y}(x, \infty)$
- (c) $F_{X,Y}(-\infty, \infty)$
- (d) $F_{X,Y}(-\infty, y)$

(e) $F_{X,Y}(\infty, y)$

5.1.3 ■ For continuous random variables X, Y with joint CDF $F_{X,Y}(x,y)$ and marginal CDFs $F_X(x)$ and $F_Y(y)$, find $P[x_1 \leq X < x_2 \cup y_1 \leq Y < y_2]$. This is the probability of the shaded “cross” region in the following diagram.



5.1.4 ■ Random variables X and Y have CDF $F_X(x)$ and $F_Y(y)$. Is $F(x, y) = F_X(x)F_Y(y)$ a valid CDF? Explain your answer.

5.1.5 ♦ In this problem, we prove Theorem 5.2.

- (a) Sketch the following events on the X, Y plane:

$$\begin{aligned} A &= \{X \leq x_1, y_1 < Y \leq y_2\}, \\ B &= \{x_1 < X \leq x_2, Y \leq y_1\}, \\ C &= \{x_1 < X \leq x_2, y_1 < Y \leq y_2\}. \end{aligned}$$

- (b) Express the probability of the events A , B , and $A \cup B \cup C$ in terms of the joint CDF $F_{X,Y}(x, y)$.
- (c) Use the observation that events A , B , and C are mutually exclusive to prove Theorem 5.2.

5.1.6 ♦ Can the following function be the joint CDF of random variables X and Y ?

$$F(x, y) = \begin{cases} 1 - e^{-(x+y)} & x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

5.2.1 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x, y) = \begin{cases} cxy & x = 1, 2, 4; \quad y = 1, 3, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[Y < X]$?
- (c) What is $P[Y > X]$?
- (d) What is $P[Y = X]$?
- (e) What is $P[Y = 3]$?

5.2.2 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x, y) = \begin{cases} c|x+y| & x = -2, 0, 2; \\ & y = -1, 0, 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?

- (b) What is $P[Y < X]$?
- (c) What is $P[Y > X]$?
- (d) What is $P[Y = X]$?
- (e) What is $P[X < 1]$?

5.2.3 ● Test two integrated circuits. In each test, the probability of rejecting the circuit is p , independent of the other test. Let X be the number of rejects (either 0 or 1) in the first test and let Y be the number of rejects in the second test. Find the joint PMF $P_{X,Y}(x, y)$.

5.2.4 ■ For two independent flips of a fair coin, let X equal the total number of tails and let Y equal the number of heads on the last flip. Find the joint PMF $P_{X,Y}(x, y)$.

5.2.5 ■ In Figure 5.2, the axes of the figures are labeled X and Y because the figures depict possible values of the random variables X and Y . However, the figure at the end of Example 5.3 depicts $P_{X,Y}(x, y)$ on axes labeled with lowercase x and y . Should those axes be labeled with the uppercase X and Y ? Hint: Reasonable arguments can be made for both views.

5.2.6 ■ As a generalization of Example 5.3, consider a test of n circuits such that each circuit is acceptable with probability p , independent of the outcome of any other test. Show that the joint PMF of X , the number of acceptable circuits, and Y , the number of acceptable circuits found before observing the first reject, is

$$\begin{aligned} P_{X,Y}(x, y) &= \begin{cases} \binom{n-y-1}{x-y} p^x (1-p)^{n-x} & 0 \leq y \leq x < n, \\ p^n & x = y = n, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Hint: For $0 \leq y \leq x < n$, show that

$$\{X = x, Y = y\} = A \cap B \cap C,$$

where

- A : The first y tests are acceptable.
 B : Test $y + 1$ is a rejection.
 C : The remaining $n - y - 1$ tests yield $x - y$ acceptable circuits

5.2.7 ■ With two minutes left in a five-minute overtime, the score is 0–0 in a Rutgers soccer match versus Villanova. (Note that the overtime is NOT *sudden-death*.) In the next-to-last minute of the game, either (1) Rutgers scores a goal with probability $p = 0.2$, (2) Villanova scores with probability $p = 0.2$, or (3) neither team scores with probability $1 - 2p = 0.6$. If neither team scores in the next-to-last minute, then in the final minute, either (1) Rutgers scores a goal with probability $q = 0.3$, (2) Villanova scores with probability $q = 0.3$, or (3) neither team scores with probability $1 - 2q = 0.4$. However, if a team scores in the next-to-last minute, the trailing team goes for broke so that in the last minute, either (1) the leading team scores with probability 0.5, or (2) the trailing team scores with probability 0.5. For the final two minutes of overtime:

- Sketch a probability tree and construct a table for $P_{R,V}(r, v)$, the joint PMF of R , the number of Rutgers goals scored, and V , the number of Villanova goals scored.
- What is the probability $P[T]$ that the overtime ends in a tie?
- What is the PMF of R , the number of goals scored by Rutgers?
- What is the PMF of G , the total number of goals scored?

5.2.8 ♦ Each test of an integrated circuit produces an acceptable circuit with probability p , independent of the outcome of the test of any other circuit. In testing n circuits, let K denote the number of circuits rejected and let X denote the number of acceptable circuits (either 0 or 1) in the last test. Find the joint PMF $P_{K,X}(k, x)$.

5.2.9 ♦ Each test of an integrated circuit produces an acceptable circuit with probability p , independent of the outcome of the test of any other circuit. In testing n circuits, let K denote the number of circuits rejected and let X denote the number of acceptable circuits that appear before the

first reject is found. Find the joint PMF $P_{K,X}(k, x)$.

5.3.1 ● Given the random variables X and Y in Problem 5.2.1, find

- The marginal PMFs $P_X(x)$ and $P_Y(y)$,
- The expected values $E[X]$ and $E[Y]$,
- The standard deviations σ_X and σ_Y .

5.3.2 ● Given the random variables X and Y in Problem 5.2.2, find

- The marginal PMFs $P_X(x)$ and $P_Y(y)$,
- The expected values $E[X]$ and $E[Y]$,
- The standard deviations σ_X and σ_Y .

5.3.3 ● For $n = 0, 1, \dots$ and $0 \leq k \leq 100$, the joint PMF of random variables N and K is

$$P_{N,K}(n, k) = \frac{100^n e^{-100}}{n!} \binom{100}{k} p^k (1-p)^{100-k}.$$

Otherwise, $P_{N,K}(n, k) = 0$. Find the marginal PMFs $P_N(n)$ and $P_K(k)$.

5.3.4 ■ Random variables X and Y have joint PMF

$$P_{X,Y}(x, y) = \begin{cases} 1/21 & x = 0, 1, 2, 3, 4, 5; \\ & y = 0, 1, \dots, x, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PMFs $P_X(x)$ and $P_Y(y)$ and the expected values $E[X]$ and $E[Y]$.

5.3.5 ■ Random variables N and K have the joint PMF

$$P_{N,K}(n, k) = \begin{cases} \frac{(1-p)^{n-1} p}{n} & k=1, \dots, n; \\ & n=1, 2, \dots \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PMFs $P_N(n)$ and $P_K(k)$.

5.3.6 ♦ Random variables N and K have the joint PMF

$$P_{N,K}(n, k) = \begin{cases} \frac{100^n e^{-100}}{(n+1)!} & k=0, 1, \dots, n; \\ & n=0, 1, \dots \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PMF $P_N(n)$. Show that the marginal PMF $P_K(k)$ satisfies $P_K(k) = P[N > k]/100$.

5.4.1 ● Random variables X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} c & x \geq 0, y \geq 0, x+y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[X \leq Y]$?
- (c) What is $P[X + Y \leq 1/2]$?

5.4.2 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cxy^2 & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the constant c .
- (b) Find $P[X > Y]$ and $P[Y < X^2]$.
- (c) Find $P[\min(X, Y) \leq 1/2]$.
- (d) Find $P[\max(X, Y) \leq 3/4]$.

5.4.3 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 6e^{-(2x+3y)} & x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find $P[X > Y]$ and $P[X + Y \leq 1]$.
- (b) Find $P[\min(X, Y) \geq 1]$.
- (c) Find $P[\max(X, Y) \leq 1]$.

5.4.4 ♦ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 8xy & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Following the method of Example 5.8, find the joint CDF $F_{X,Y}(x,y)$.

5.5.1 ● Random variables X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/2 & -1 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Sketch the region of nonzero probability and answer the following questions.

- (a) What is $P[X > 0]$?
- (b) What is $f_X(x)$?
- (c) What is $E[X]$?

5.5.2 ● Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cx & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the constant c .
- (b) Find the marginal PDF $f_X(x)$.
- (c) Are X and Y independent? Justify your answer.

5.5.3 ● X and Y are random variables with the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & x+y \leq 1, x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the marginal PDF $f_X(x)$?
- (b) What is the marginal PDF $f_Y(y)$?

5.5.4 ■ Over the circle $X^2 + Y^2 \leq r^2$, random variables X and Y have the uniform PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/(\pi r^2) & x^2 + y^2 \leq r^2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the marginal PDF $f_X(x)$?
- (b) What is the marginal PDF $f_Y(y)$?

5.5.5 ■ X and Y are random variables with the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 5x^2/2 & -1 \leq x \leq 1, \\ 0 & 0 \leq y \leq x^2 \\ & \text{otherwise.} \end{cases}$$

- (a) What is the marginal PDF $f_X(x)$?
- (b) What is the marginal PDF $f_Y(y)$?

5.5.6 ■ Over the circle $X^2 + Y^2 \leq r^2$, random variables X and Y have the PDF

$$f_{X,Y}(x,y) = \begin{cases} 2|xy|/r^4 & x^2 + y^2 \leq r^2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the marginal PDF $f_X(x)$?
 (b) What is the marginal PDF $f_Y(y)$?

5.5.7 ■ For a random variable X , let $Y = aX + b$. Show that if $a > 0$ then $\rho_{X,Y} = 1$. Also show that if $a < 0$, then $\rho_{X,Y} = -1$.

5.5.8 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} (x+y)/3 & 0 \leq x \leq 1; \\ & 0 \leq y \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.
 (b) What are $E[X]$ and $\text{Var}[X]$?
 (c) What are $E[Y]$ and $\text{Var}[Y]$?

5.5.9 ♦ Random variables X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cy & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Draw the region of nonzero probability.
 (b) What is the value of the constant c ?
 (c) What is $F_X(x)$?
 (d) What is $F_Y(y)$?
 (e) What is $P[Y \leq X/2]$?

5.6.1 ● An ice cream company needs to order ingredients from its suppliers. Depending on the size of the order, the weight of the shipment can be either

- 1 kg for a small order,
 2 kg for a big order.

The company has three different suppliers. The vanilla supplier is 20 miles away. The chocolate supplier is 100 miles away. The

strawberry supplier is 300 miles away. An experiment consists of monitoring an order and observing W , the weight of the order, and D , the distance the shipment must be sent. The following probability model describes the experiment:

	van.	choc.	straw.
small	0.2	0.2	0.2
big	0.1	0.2	0.1

- (a) What is the joint PMF $P_{W,D}(w,d)$ of the weight and the distance?
 (b) Find the expected shipping distance $E[D]$.
 (c) Are W and D independent?

5.6.2 ● A company receives shipments from two factories. Depending on the size of the order, a shipment can be in

- 1 box for a small order,
 2 boxes for a medium order,
 3 boxes for a large order.

The company has two different suppliers. Factory Q is 60 miles from the company. Factory R is 180 miles from the company.

An experiment consists of monitoring a shipment and observing B , the number of boxes, and M , the number of miles the shipment travels. The following probability model describes the experiment:

	Factory Q	Factory R
small order	0.3	0.2
medium order	0.1	0.2
large order	0.1	0.1

- (a) Find $P_{B,M}(b,m)$, the joint PMF of the number of boxes and the distance.
 (b) What is $E[B]$, the expected number of boxes?
 (c) Are B and M independent?

5.6.3 ● Observe 100 independent flips of a fair coin. Let X equal the number of heads in the first 75 flips. Let Y equal the number of heads in the remaining 25 flips. Find $P_X(x)$ and $P_Y(y)$. Are X and Y independent? Find $P_{X,Y}(x,y)$.

5.6.4 ● Observe independent flips of a fair coin until heads occurs twice. Let X_1 equal the number of flips up to and including the

first H . Let X_2 equal the number of additional flips up to and including the second H . What are $P_{X_1}(x_1)$ and $P_{X_2}(x_2)$. Are X_1 and X_2 independent? Find $P_{X_1, X_2}(x_1, x_2)$.

5.6.5 ● X is the continuous uniform $(0, 2)$ random variable. Y has the continuous uniform $(0, 5)$ PDF, independent of X . What is the joint PDF $f_{X,Y}(x, y)$?

5.6.6 ● X_1 and X_2 are independent random variables such that X_i has PDF

$$f_{X_i}(x) = \begin{cases} \lambda_i e^{-\lambda_i x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

What is $P[X_2 < X_1]$?

5.6.7 ■ In terms of a positive constant k , random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} k + 3x^2 & -1/2 \leq x \leq 1/2, \\ & -1/2 \leq y \leq 1/2, \\ 0 & \text{otherwise.} \end{cases}$$

- What is k ?
- What is the marginal PDF of X ?
- What is the marginal PDF of Y ?
- Are X and Y independent?

5.6.8 ■ X_1 and X_2 are independent, identically distributed random variables with PDF

$$f_X(x) = \begin{cases} x/2 & 0 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

- Find the CDF, $F_X(x)$.
- What is $P[X_1 \leq 1, X_2 \leq 1]$, the probability that X_1 and X_2 are both less than or equal to 1?
- Let $W = \max(X_1, X_2)$. What is $F_W(1)$, the CDF of W evaluated at $w = 1$?
- Find the CDF $F_W(w)$.

5.6.9 ♦ Prove that random variables X and Y are independent if and only if

$$F_{X,Y}(x, y) = F_X(x) F_Y(y).$$

5.7.1 ● Continuing Problem 5.6.1, the price per kilogram for shipping the order is one cent per mile. C cents is the shipping cost of one order. What is $E[C]$?

5.7.2 ● Continuing Problem 5.6.2, the price per mile of shipping each box is one cent per mile the box travels. C cents is the price of one shipment. What is $E[C]$, the expected price of one shipment?

5.7.3 ● A random ECE sophomore has height X (rounded to the nearest foot) and GPA Y (rounded to the nearest integer). These random variables have joint PMF

$P_{X,Y}(x, y)$	$y = 1$	$y = 2$	$y = 3$	$y = 4$
$x = 5$	0.05	0.1	0.2	0.05
$x = 6$	0.1	0.1	0.3	0.1

Find $E[X + Y]$ and $\text{Var}[X + Y]$.

5.7.4 ● X and Y are independent, identically distributed random variables with PMF

$$P_X(k) = P_Y(k) = \begin{cases} 3/4 & k = 0, \\ 1/4 & k = 20, \\ 0 & \text{otherwise.} \end{cases}$$

Find the following quantities:

$$E[X], \quad \text{Var}[X], \\ E[X + Y], \quad \text{Var}[X + Y], \quad E[XY2^{XY}].$$

5.7.5 ● X and Y are random variables with $E[X] = E[Y] = 0$ and $\text{Var}[X] = 1$, $\text{Var}[Y] = 4$ and correlation coefficient $\rho = 1/2$. Find $\text{Var}[X + Y]$.

5.7.6 ● X and Y are random variables such that X has expected value $\mu_X = 0$ and standard deviation $\sigma_X = 3$ while Y has expected value $\mu_Y = 1$ and standard deviation $\sigma_Y = 4$. In addition, X and Y have covariance $\text{Cov}[X, Y] = -3$. Find the expected value and variance of $W = 2X + 2Y$.

5.7.7 ● Observe independent flips of a fair coin until heads occurs twice. Let X_1 equal the number of flips up to and including the first H . Let X_2 equal the number of additional flips up to and including the sec-

and H . Let $Y = X_1 - X_2$. Find $E[Y]$ and $\text{Var}[Y]$. Hint: Don't try to find $P_Y(y)$.

5.7.8 ● X_1 and X_2 are independent identically distributed random variables with expected value $E[X]$ and variance $\text{Var}[X]$.

- (a) What is $E[X_1 - X_2]$?
- (b) What is $\text{Var}[X_1 - X_2]$?

5.7.9 ■ X and Y are identically distributed random variables with $E[X] = E[Y] = 0$ and covariance $\text{Cov}[X, Y] = 3$ and correlation coefficient $\rho_{X,Y} = 1/2$. For nonzero constants a and b , $U = aX$ and $V = bY$.

- (a) Find $\text{Cov}[U, V]$.
- (b) Find the correlation coefficient $\rho_{U,V}$.
- (c) Let $W = U + V$. For what values of a and b are X and W uncorrelated?

5.7.10 ■ True or False: For identically distributed random variables Y_1 and Y_2 with $E[Y_1] = E[Y_2] = 0$, $\text{Var}[Y_1 + Y_2] \geq \text{Var}[Y_1]$.

5.7.11 ■ X and Y are random variables with $E[X] = E[Y] = 0$ such that X has standard deviation $\sigma_X = 2$ while Y has standard deviation $\sigma_Y = 4$.

- (a) For $V = X - Y$, what are the smallest and largest possible values of $\text{Var}[V]$?
- (b) For $W = X - 2Y$, what are the smallest and largest possible values of $\text{Var}[W]$?

5.7.12 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 4xy & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What are $E[X]$ and $\text{Var}[X]$?
- (b) What are $E[Y]$ and $\text{Var}[Y]$?
- (c) What is $\text{Cov}[X, Y]$?
- (d) What is $E[X + Y]$?
- (e) What is $\text{Var}[X + Y]$?

5.7.13 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 5x^2/2 & -1 \leq x \leq 1; \\ & 0 \leq y \leq x^2, \\ 0 & \text{otherwise.} \end{cases}$$

Answer the following questions.

- (a) What are $E[X]$ and $\text{Var}[X]$?
- (b) What are $E[Y]$ and $\text{Var}[Y]$?
- (c) What is $\text{Cov}[X, Y]$?
- (d) What is $E[X + Y]$?
- (e) What is $\text{Var}[X + Y]$?

5.7.14 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What are $E[X]$ and $\text{Var}[X]$?
- (b) What are $E[Y]$ and $\text{Var}[Y]$?
- (c) What is $\text{Cov}[X, Y]$?
- (d) What is $E[X + Y]$?
- (e) What is $\text{Var}[X + Y]$?

5.7.15 ■ A transmitter sends a signal X and a receiver makes the observation $Y = X + Z$, where Z is a receiver noise that is independent of X and $E[X] = E[Z] = 0$. Since the average power of the signal is $E[X^2]$ and the average power of the noise is $E[Z^2]$, a quality measure for the received signal is the signal-to-noise ratio

$$\Gamma = \frac{E[X^2]}{E[Z^2]}.$$

How is Γ related to the correlation coefficient $\rho_{X,Y}$?

5.8.1 ● X and Z are independent random variables with $E[X] = E[Z] = 0$ and variance $\text{Var}[X] = 1$ and $\text{Var}[Z] = 16$. Let $Y = X + Z$. Find the correlation coefficient ρ of X and Y . Are X and Y independent?

5.8.2 ● For the random variables X and Y in Problem 5.2.1, find

- (a) The expected value of $W = Y/X$,
- (b) The correlation, $r_{X,Y} = E[XY]$,
- (c) The covariance, $\text{Cov}[X, Y]$,

- (d) The correlation coefficient, $\rho_{X,Y}$,
 (e) The variance of $X + Y$, $\text{Var}[X + Y]$.

(Refer to the results of Problem 5.3.1 to answer some of these questions.)

5.8.3 For the random variables X and Y in Problem 5.2.2 find

- (a) The expected value of $W = 2^{XY}$,
 (b) The correlation, $r_{X,Y} = E[XY]$,
 (c) The covariance, $\text{Cov}[X, Y]$,
 (d) The correlation coefficient, $\rho_{X,Y}$,
 (e) The variance of $X + Y$, $\text{Var}[X + Y]$.

(Refer to the results of Problem 5.3.2 to answer some of these questions.)

5.8.4 Let H and B be the random variables in Quiz 5.3. Find $r_{H,B}$ and $\text{Cov}[H, B]$.

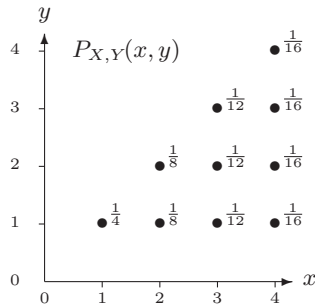
5.8.5 X and Y are independent random variables with PDFs

$$f_X(x) = \begin{cases} \frac{1}{3}e^{-x/3} & x \geq 0, \\ 0 & \text{otherwise,} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2}e^{-y/2} & y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the correlation $r_{X,Y}$.
 (b) Find the covariance $\text{Cov}[X, Y]$.

5.8.6 The random variables X and Y have joint PMF



Find

- (a) The expected values $E[X]$ and $E[Y]$,
 (b) The variances $\text{Var}[X]$ and $\text{Var}[Y]$,

- (c) The correlation, $r_{X,Y} = E[XY]$,
 (d) The covariance, $\text{Cov}[X, Y]$,
 (e) The correlation coefficient, $\rho_{X,Y}$.

5.8.7 For X and Y with PMF $P_{X,Y}(x, y)$ given in Problem 5.8.6, let $W = \min(X, Y)$ and $V = \max(X, Y)$. Find

- (a) The expected values, $E[W]$ and $E[V]$,
 (b) The variances, $\text{Var}[W]$ and $\text{Var}[V]$,
 (c) The correlation, $r_{W,V}$,
 (d) The covariance, $\text{Cov}[W, V]$,
 (e) The correlation coefficient, $\rho_{W,V}$.

5.8.8 Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 1/2 & -1 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find $r_{X,Y}$ and $E[e^{X+Y}]$.

5.8.9 This problem outlines a proof of Theorem 5.13.

- (a) Show that

$$\hat{X} - E[\hat{X}] = a(X - E[X]),$$

$$\hat{Y} - E[\hat{Y}] = c(Y - E[Y]).$$

- (b) Use part (a) to show that

$$\text{Cov}[\hat{X}, \hat{Y}] = ac \text{Cov}[X, Y].$$

- (c) Show that $\text{Var}[\hat{X}] = a^2 \text{Var}[X]$ and $\text{Var}[\hat{Y}] = c^2 \text{Var}[Y]$.

- (d) Combine parts (b) and (c) to relate $\rho_{\hat{X}, \hat{Y}}$ and $\rho_{X,Y}$.

5.8.10 Random variables N and K have the joint PMF

$$P_{N,K}(n, k) = \begin{cases} (1-p)^{n-1}p/n & k = 1, \dots, n; \\ & n = 1, 2, \dots, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PMF $P_N(n)$ and the expected values $E[N]$, $\text{Var}[N]$, $E[N^2]$, $E[K]$, $\text{Var}[K]$, $E[N + K]$, $r_{N,K}$, $\text{Cov}[N, K]$.

5.9.1 Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = ce^{-(x^2/8) - (y^2/18)}.$$

What is the constant c ? Are X and Y independent?

5.9.2 X is the Gaussian ($\mu = 1, \sigma = 2$) random variable. Y is the Gaussian ($\mu = 2, \sigma = 4$) random variable. X and Y are independent.

- (a) What is the PDF of $V = X + Y$?
- (b) What is the PDF of $W = 3X + 2Y$?

5.9.3 TRUE OR FALSE: X_1 and X_2 are bivariate Gaussian random variables. For any constant y , there exists a constant a such that $P[X_1 + aX_2 \leq y] = 1/2$.

5.9.4 X_1 and X_2 are identically distributed Gaussian $(0, 1)$ random variables. Moreover, they are jointly Gaussian. Under what conditions are X_1 , X_2 and $X_1 + X_2$ identically distributed?

5.9.5 Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = ce^{-(2x^2 - 4xy + 4y^2)}.$$

- (a) What are $E[X]$ and $E[Y]$?
- (b) Find the correlation coefficient $\rho_{X,Y}$.
- (c) What are $\text{Var}[X]$ and $\text{Var}[Y]$?
- (d) What is the constant c ?
- (e) Are X and Y independent?

5.9.6 An archer shoots an arrow at a circular target of radius 50 cm. The arrow pierces the target at a random position (X, Y) , measured in centimeters from the center of the disk at position $(X, Y) = (0, 0)$. The bullseye is a solid black circle of radius 2 cm, at the center of the target. Calculate the probability $P[B]$ of the event that the archer hits the bullseye under each of the following models:

- (a) X and Y are iid continuous uniform $(-50, 50)$ random variables.

(b) The PDF $f_{X,Y}(x, y)$ is uniform over the 50 cm circular target.

(c) X and Y are iid Gaussian ($\mu = 0, \sigma = 10$) random variables.

5.9.7 A person's white blood cell (WBC) count W (measured in thousands of cells per microliter of blood) and body temperature T (in degrees Celsius) can be modeled as bivariate Gaussian random variables such that W is Gaussian $(7, 2)$ and T is Gaussian $(37, 1)$. To determine whether a person is sick, first the person's temperature T is measured. If $T > 38$, then the person's WBC count is measured. If $W > 10$, the person is declared ill (event I).

- (a) Suppose W and T are uncorrelated. What is $P[I]$? Hint: Draw a tree diagram for the experiment.
- (b) Now suppose W and T have correlation coefficient $\rho_{W,T} = 1/\sqrt{2}$. Find the conditional probability $P[I|T = t]$ that a person is declared ill given that the person's temperature is $T = t$.

5.9.8 Suppose your grade in a probability course depends on your exam scores X_1 and X_2 . The professor, a fan of probability, releases exam scores in a normalized fashion such that X_1 and X_2 are iid Gaussian ($\mu = 0, \sigma = \sqrt{2}$) random variables. Your semester average is $X = 0.5(X_1 + X_2)$.

- (a) You earn an A grade if $X > 1$. What is $P[A]$?
- (b) To improve his SIRS (Student Instructional Rating Service) score, the professor decides he should award more A 's. Now you get an A if $\max(X_1, X_2) > 1$. What is $P[A]$ now?
- (c) The professor found out he is unpopular at ratemyprofessor.com and decides to award an A if either $X > 1$ or $\max(X_1, X_2) > 1$. Now what is $P[A]$?
- (d) Under criticism of grade inflation from the department chair, the professor adopts a new policy. An A is awarded if $\max(X_1, X_2) > 1$ and $\min(X_1, X_2) > 0$. Now what is $P[A]$?

5.9.9 ♦ Your course grade depends on two test scores: X_1 and X_2 . Your score X_i on test i is Gaussian ($\mu = 74, \sigma = 16$) random variable, independent of any other test score.

- (a) With equal weighting, grades are determined by $Y = X_1/2 + X_2/2$. You earn an A if $Y \geq 90$. What is $P[A] = P[Y \geq 90]$?
- (b) A student asks the professor to choose a weight factor w , $0 \leq w \leq 1$, such that

$$Y = wX_1 + (1 - w)X_2.$$

Find $P[A]$ as a function of the weight w . What value or values of w maximize $P[A] = P[Y \geq 90]$?

- (c) A different student proposes that the better exam is the one that should count and that grades should be based on $M = \max(X_1, X_2)$. In a fit of generosity, the professor agrees! Now what is $P[A] = P[M > 90]$?
- (d) How generous was the professor? In a class of 100 students, what is the expected increase in the number of A's awarded?

5.9.10 ♦ Under what conditions on the constants a , b , c , and d is

$$f(x, y) = de^{-(a^2x^2 + bxy + c^2y^2)}$$

a joint Gaussian PDF?

5.9.11 ♦ Show that the joint Gaussian PDF $f_{X,Y}(x, y)$ given by Definition 5.10 satisfies

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dx \, dy = 1.$$

Hint: Use Equation (5.68) and the result of Problem 4.6.13.

5.9.12 ♦ Random variables X_1 and X_2 are independent identical Gaussian $(0, 1)$ random variables. Let

$$Y_1 = X_1 \operatorname{sgn}(X_2), \quad Y_2 = X_2 \operatorname{sgn}(X_1),$$

where

$$\operatorname{sgn}(x) = \begin{cases} 1 & x > 0, \\ -1 & x \leq 0. \end{cases}$$

- (a) Find the CDF $F_{Y_1}(y_1)$ in terms of the $\Phi(\cdot)$ function.
- (b) Show that Y_1 and Y_2 are both Gaussian random variables.
- (c) Are Y_1 and Y_2 bivariate Gaussian random variables?

5.10.1 ● Every laptop returned to a repair center is classified according its needed repairs: (1) LCD screen, (2) motherboard, (3) keyboard, or (4) other. A random broken laptop needs a type i repair with probability $p_i = 2^{4-i}/15$. Let N_i equal the number of type i broken laptops returned on a day in which four laptops are returned.

- (a) Find the joint PMF of N_1, N_2, N_3, N_4 .
- (b) What is the probability that two laptops require LCD repairs?
- (c) What is the probability that more laptops require motherboard repairs than keyboard repairs?

5.10.2 ● When ordering a personal computer, a customer can add the following features to the basic configuration: (1) additional memory, (2) flat panel display, (3) professional software, and (4) wireless modem. A random computer order has feature i with probability $p_i = 2^{-i}$ independent of other features. In an hour in which three computers are ordered, let N_i equal the number of computers with feature i .

- (a) Find the joint PMF

$$P_{N_1, N_2, N_3, N_4}(n_1, n_2, n_3, n_4).$$

- (b) What is the probability of selling a computer with no additional features?
- (c) What is the probability of selling a computer with at least three additional features?

5.10.3 The random variables $X_1, \dots, X_n$ have the joint PDF

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \begin{cases} 1 & 0 \leq x_i \leq 1; \\ & i = 1, \dots, n, \\ 0 & \text{otherwise.} \end{cases}$$

Find

- (a) The joint CDF, $F_{X_1, \dots, X_n}(x_1, \dots, x_n)$,
- (b) $P[\min(X_1, X_2, X_3) \leq 3/4]$.

5.10.4 Are N_1, N_2, N_3, N_4 in Problem 5.10.1 independent?

5.10.5 In a compressed data file of 10,000 bytes, each byte is equally likely to be any one of 256 possible characters $b_0, \dots, b_{255}$ independent of any other byte. If N_i is the number of times b_i appears in the file, find the joint PMF of $N_0, \dots, N_{255}$. Also, what is the joint PMF of N_0 and N_1 ?

5.10.6 In Example 5.22, we derived the joint PMF of the the number of pages in each of four downloads:

$$P_{X,Y,Z}(x, y, z) = \begin{pmatrix} 4 \\ x, y, z \end{pmatrix} \frac{1}{3^x} \frac{1}{2^y} \frac{1}{6^z}.$$

- (a) In a group of four downloads, what is the PMF of the number of 3-page documents?
- (b) In a group of four downloads, what is the expected number of 3-page documents?
- (c) Given that there are two 3-page documents in a group of four, what is the joint PMF of the number of 1-page documents and the number of 2-page documents?
- (d) Given that there are two 3-page documents in a group of four, what is the expected number of 1-page documents?
- (e) In a group of four downloads, what is the joint PMF of the number of 1-page documents and the number of 2-page documents?

5.10.7 X_1, X_2, X_3 are iid exponential (λ) random variables. Find:

- (a) the PDF of $V = \min(X_1, X_2, X_3)$,
- (b) the PDF of $W = \max(X_1, X_2, X_3)$.

5.10.8 In a race of 10 sailboats, the finishing times of all boats are iid Gaussian random variables with expected value 35 minutes and standard deviation 5 minutes.

- (a) What is the probability that the winning boat will finish the race in less than 25 minutes?
- (b) What is the probability that the last boat will cross the finish line in more than 50 minutes?
- (c) Given this model, what is the probability that a boat will finish before it starts (negative finishing time)?

5.10.9 Random variables $X_1, X_2, \dots, X_n$ are iid; each X_j has CDF $F_X(x)$ and PDF $f_X(x)$. Consider

$$L_n = \min(X_1, \dots, X_n) \\ U_n = \max(X_1, \dots, X_n).$$

In terms of $F_X(x)$ and/or $f_X(x)$:

- (a) Find the CDF $F_{U_n}(u)$.
- (b) Find the CDF $F_{L_n}(l)$.
- (c) Find the joint CDF $F_{L_n, U_n}(l, u)$.

5.10.10 Suppose you have n suitcases and suitcase i holds X_i dollars where $X_1, X_2, \dots, X_n$ are iid continuous uniform $(0, m)$ random variables. (Think of a number like one million for the symbol m .) Unfortunately, you don't know X_i until you open suitcase i .

Suppose you can open the suitcases one by one, starting with suitcase n and going down to suitcase 1. After opening suitcase i , you can either accept or reject X_i dollars. If you accept suitcase i , the game ends. If you reject, then you get to choose only from the still unopened suitcases.

What should you do? Perhaps it is not so obvious? In fact, you can decide before the game on a policy, a set of rules to follow. We will specify a policy by a vector $(\tau_1, \dots, \tau_n)$ of threshold parameters.

- After opening suitcase i , you accept the amount X_i if $X_i \geq \tau_i$.
 - Otherwise, you reject suitcase i and open suitcase $i - 1$.
 - If you have rejected suitcases n down through 2, then you must accept the amount X_1 in suitcase 1. Thus the threshold $\tau_1 = 0$ since you never reject the amount in the last suitcase.
- (a) Suppose you reject suitcases n through $i + 1$, but then you accept suitcase i . Find $E[X_i | X_i \geq \tau_i]$.
- (b) Let W_k denote your reward given that there are k unopened suitcases remaining. What is $E[W_1]$?
- (c) As a function of τ_k , find a recursive relationship for $E[W_k]$ in terms of τ_k and $E[W_{k-1}]$.
- (d) For $n = 4$ suitcases, find the policy $(\tau_1^*, \dots, \tau_4^*)$, that maximizes $E[W_4]$.

5.10.11 ♦♦ Given the set $\{U_1, \dots, U_n\}$ of iid uniform $(0, T)$ random variables, we define

$$X_k = \text{small}_k(U_1, \dots, U_n)$$

as the k th “smallest” element of the set. That is, X_1 is the minimum element, X_2 is the second smallest, and so on, up to X_n , which is the maximum element of $\{U_1, \dots, U_n\}$. Note that $X_1, \dots, X_n$ are known as the *order statistics* of $U_1, \dots, U_n$. Prove that

$$\begin{aligned} f_{X_1, \dots, X_n}(x_1, \dots, x_n) \\ = \begin{cases} n!/T^n & 0 \leq x_1 < \dots < x_n \leq T, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

5.11.1 ● For random variables X and Y in Example 5.26, use MATLAB to generate a list of the form

$$\begin{array}{ccc} x_1 & y_1 & P_{X,Y}(x_1, y_1) \\ x_2 & y_2 & P_{X,Y}(x_2, y_2) \\ \vdots & \vdots & \vdots \end{array}$$

that includes all possible pairs (x, y) .

5.11.2 ● For random variables X and Y in Example 5.26, use MATLAB to calculate $E[X]$, $E[Y]$, the correlation $E[XY]$, and the covariance $\text{Cov}[X, Y]$.

5.11.3 ● You generate random variable $W = \mathbf{W}$ by typing `W=sum(4*randn(1,2))` in a MATLAB Command window. What is $\text{Var}[W]$?

5.11.4 ● Write `trianglecdfplot.m`, a script that graphs $F_{X,Y}(x, y)$ of Figure 5.4.

5.11.5 ■ Problem 5.2.6 extended Example 5.3 to a test of n circuits and identified the joint PDF of X , the number of acceptable circuits, and Y , the number of successful tests before the first reject. Write a MATLAB function

$$[\mathbf{SX}, \mathbf{SY}, \mathbf{PXY}] = \text{circuits}(n, p)$$

that generates the sample space grid for the n circuit test. Check your answer against Equation (5.11) for the $p = 0.9$ and $n = 2$ case. For $p = 0.9$ and $n = 50$, calculate the correlation coefficient $\rho_{X,Y}$.